



**NANYANG
TECHNOLOGICAL
UNIVERSITY**

SINGAPORE

Treebanking a Corpus of Learner English

by

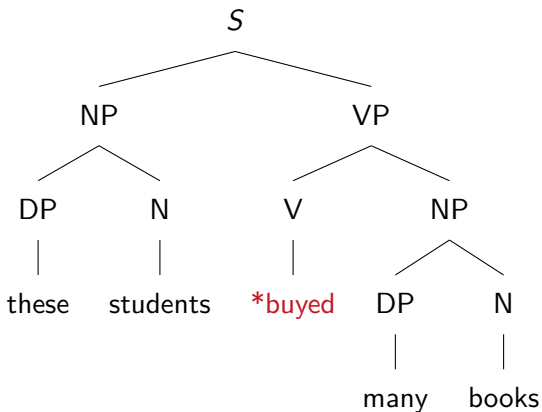
Luis Morgado da Costa, Francis Bond, Dan Flickinger

Background

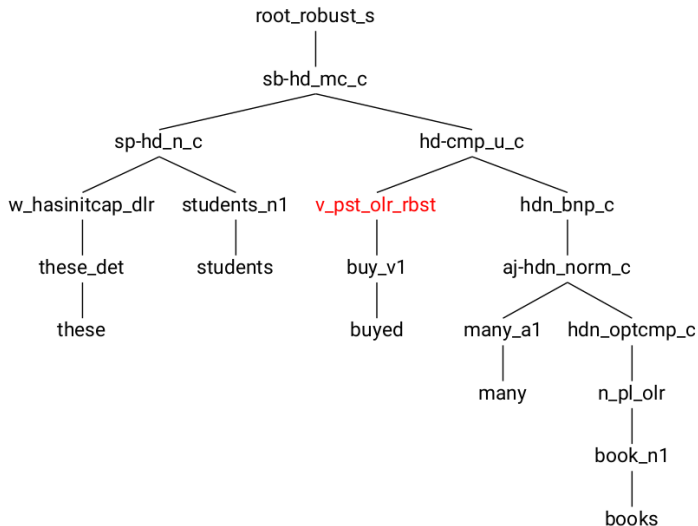
- ❖ We have a good collection of treebanks for the ERG:
mainly edited text: news, wiki, stories; some speech; some
phenomena-based test-suites
- ❖ but we don't have any treebanks for mal-rules
 - ❖ Treebank the NTU Corpus of Learner Errors
 - ❖ FFTB online for multiple annotators
 - ❖ Itdb and wiki for documentation
 - ❖ train by doing and discussing

Mal-Rule (Inflectional Rule)

* These students bought many books.



ERG Structure



Original Design:

- ❖ 2 Grammars (ERG + (ERG+mal))
- ❖ ERG mal-rules only checked if ERG rejected the sentence
- ❖ Inherent flexibility of ERG means many errors treated as wierd interpretations

Goal:

- ❖ Change to 1 Grammar process: ERG+mal
- ❖ Need to train a parsing model with mal-rules

- ❖ NTU Corpus of Learner English (NTUCLE)
 - ❖ 1st year engineering course assignments
 - ❖ 500-800 words per document
 - ❖ describe and propose a solution to everyday problems
- ❖ 180 documents tagged with errors by writing instructors
 - ❖ ~9,500 sents, ~120,00 words
- ❖ ~800 documents (CC-BY) (growing every semester)
 - ❖ filtered for meaning text blocks
 - ❖ ~25,200 sents, ~545,000 words
- ❖ 1,100+ resubmissions (CC-BY)
 - ❖ iterative corrections of the same documents

Treebanking with the ERG+mal

- ❖ Hosting 4 instances of FFTB on a server
 - ❖ recompiled to allow a fixed-port
 - ❖ 500 item profile takes 2G memory
 - ❖ memory usage goes up steadily over time
 - ❖ periodically restart
- ❖ Training 4 NTU LMS students (500 sentences)
- ❖ Using LTDB as base documentation, with wiki for treebanking guidelines

Goal:

- ❖ Tag at least 5,000 sentences
- ❖ Have enough overlap to report inter-annotator agreement

Treebanking Goal

Profiles				Annotators				Overlap	Training
Name	Size	Dan	Siew Yeng	Annotator1	Annotator2	Annotator3	Annotator4		
ntucle000a	216	216	216	216	216	216	216	216	Training
ntucle000b	284	-	284	284	284	284	284	284	
ntucle001	200	-	-	200	200	-	-	200	
ntucle002	200	-	-	-	-	200	200	200	
ntucle003	200	-	-	200	-	-	-	-	
ntucle004	200	-	-	-	200	-	-	-	
ntucle005	200	-	-	-	-	200	-	-	
ntucle006	200	-	-	-	-	-	200	-	
ntucle007	200	-	-	200	-	-	-	-	
ntucle008	200	-	-	-	200	-	-	-	
ntucle009	200	-	-	-	-	200	-	-	
ntucle010	200	-	-	-	-	-	200	-	
ntucle011	200	-	-	200	-	200	-	200	
ntucle012	200	-	-	-	200	-	200	200	
ntucle013	200	-	-	200	-	-	-	-	
ntucle014	200	-	-	-	200	-	-	-	
ntucle015	200	-	-	-	-	200	-	-	
ntucle016	200	-	-	-	-	-	200	-	
ntucle017	200	-	-	200	-	-	-	-	
ntucle018	200	-	-	-	200	-	-	-	
ntucle019	200	-	-	-	-	200	-	-	
ntucle020	200	-	-	-	-	-	200	-	
ntucle021	200	-	-	200	-	-	200	200	
ntucle022	200	-	-	-	200	200	-	200	
ntucle023	200	-	-	200	-	-	-	-	
ntucle024	200	-	-	-	200	-	-	-	
ntucle025	200	-	-	-	-	200	-	-	
ntucle026	200	-	-	-	-	-	200	-	
ntucle027	200	-	-	200	-	-	-	-	
ntucle028	200	-	-	-	200	-	-	-	
ntucle029	200	-	-	-	-	200	-	-	
ntucle030	200	-	-	-	-	-	200	-	
Total	6500			2300	2300	2300	2300	1700	26.15%

Adjudication Training					
ntucle000b		Genevieve	Kristianne	Nathanael	JiaQi
	20011000010 - 20014000130 (95 sents)	95	95	-	-
		-	-	95	95
		2HG: Genevieve-Kristianne		2HG: Nathanael-JiaQi	
		95		95	
	20014000140 - 20018000210 (95 sents)	95	-	95	-
		-	95	-	95
		2HG: Genevieve-Nathanael		2HG: Kristianne-JiaQi	
		95		95	
	20018000220 - 20022000120 (94 sents)	94	-	-	94
		-	94	94	-
		2HG: Genevieve-JiaQi		2HG: Kristianne-Nathanael	
		94		94	

- ❖ FFTB adapted well to remote use
although with some effort
- ❖ Linguistic Type Documentation essential for non-Dan
treebankers
we will try to improve it as we go along
- ❖ Still hard to get perfect agreement
should try to add some automatic consistency checks
- ❖ Treebank still under construction
We hope to have it ready in a month or so, ...

Thank you!